# Sivakorn Khundilokrattaya

Computer Engineering Student • Aspiring Data Scientist & ML Engineer
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## EDUCATION

**Srinakharinwirot University (SWU)** 2024 – Present (Expected 2026)
*Bachelor of Engineering in Computer Engineering* *Bangkok, Thailand*

- Specialization: Statistical Machine Learning, Financial Risk Modeling, and Distributed Data Engineering.
- Relevant Coursework: Machine Learning, Database Systems, Operating Systems, Algorithms, Software Engineering.

## TECHNICAL SKILLS & COMPETENCIES

**Programming:** Python (Expert), SQL (PostgreSQL, MySQL), C, C++, Java, R, Bash scripting
**Machine Learning & XAI:** LightGBM, XGBoost, Scikit-Learn, PyG (R-GCN), ONNX Runtime, SHAP, LIME, OptBinning, Pandas, NumPy
**Data Eng & DevOps:** Apache Spark, Apache Airflow, Docker, FastAPI, PostgreSQL, Git, CI/CD pipelines
**Languages & Ethics:** Thai (Native), English (B1+ Professional), Japanese (Elementary) • Clean OOP, PEP-8

## FEATURED ENGINEERING PROJECTS

**FlowSense & TrustGraph – FinTech Cashflow & Graph Fraud Defense** 2026
*PyTorch Geometric (R-GCN), LightGBM, ONNX Runtime, FastAPI, React 18 • KBTG Kampus Hackathon 2026*

- Built R-GCN graph neural network embeddings with ONNX Runtime sub-12ms P99 inference for real-time mule account screening.
- Deployed 30-day liquidity forecasting with normal-dip alert suppression and surplus-only micro-savings sweeps with 1-tap undo.

**Shopee Smart COD Reliability & Dynamic Decisioning Engine** 2026
*Python, Scikit-Learn, Random Forest, FastAPI, Vanilla JS • Shopee Ultimate Case Challenge (SUCC 2026)*

- Trained calibrated Random Forest pipeline (0.8629 ROC-AUC, 0.3722 PR-AUC) on 50,000 synthetic orders to predict COD delivery failures.
- Designed 4-tier checkout risk policy (Grade A 1-click COD to Grade D 30% deposit) and dynamic scorecard (+8/-20 pts feedback).

**Enterprise Credit Risk Scoring & Explainable AI (XAI)** 2026
*Python, LightGBM, SHAP, FastAPI, Docker • Basel II/III Credit Decisioning Engine*

- Constructed WoE and Information Value (IV) transformation pipelines to discretize financial attributes into monotonic risk bins.
- Integrated SHAP TreeExplainer for ECOA/FCRA Adverse Action notices; containerized FastAPI microservice with <50ms PD scoring.

**Long Overdue Debtor (LOD) Classification System** Dec 2025
*LightGBM, XGBoost, Scikit-Learn • Aihack Thailand 2025 National Finalist*

- Formulated temporal velocity ratios and credit exhaustion slopes across ~40,000 records; achieved 0.894 ROC-AUC via 5-Fold CV.

## COMPETITIVE HACKATHONS & INDUSTRY CHALLENGES

**KBTG Kampus Hackathon 2026 – Track 2 Lead (Data Science & Intelligence)** 2026
*Kasikorn Business-Technology Group (KBTG) • Status: Advanced to Next Round* *Bangkok, Thailand*

- Formulated FlowSense cashflow forecasting and TrustGraph pre-transfer fraud defense, projecting 1.2B–2.0B THB CASA deposit impact.

**Shopee Ultimate Case Challenge (SUCC) 2026 – Lead Data Scientist & Product Architect** 2026
*Shopee Thailand Operations Strategy Track • Interactive Mobile Shopping Simulator* *Bangkok, Thailand*

- Validated 10.6× COD failure risk concentration ($z=24.42$, $p < 0.001$), modeling 3.58M THB monthly savings from 58% RTO reduction.

**AIRA & AIFUL / Aihack Thailand 2025 – National Finalist** Dec 2025
*Machine Learning & Financial Risk Modeling Lead • Chulalongkorn Business School × ProbSpace* *Bangkok, Thailand*

- Handled 1:12 default imbalance using SMOTE-Tomek; pitched default risk reduction vs. interest margin profitability to C-level executives.

**True Innovation Launchpad & GISTDA Spatial Resilience Hackathons** 2026
*AI Lead (True Healthcare Frailty Prevention APIs) • Spatial Lead (GISTDA Rain-to-Flood 30m Mapping)* *Bangkok, Thailand*

## CERTIFICATIONS & STANDARDIZED ASSESSMENTS

**SKILLKAMP by KBTG:** Data Analyst (STAMP) Score: **97/150** (Benchmark: 90) • Digital Marketer Score: **98/150** (Benchmark: 94)
**Oxford Placement Test:** CEFR B1+ Professional (Score: **51/120** – Listening: 51.12, Grammar: 50.88; AUA / Oxford)